Day(38)

(ii) For loop OR For. . . else loop:-

syntax① for varname in iterable object:
Indentation block of stmts
other statements in program

syntax② for varname in iterable object:
Indentation block of stmts.
else:
else block of stmts.
other statements in program

=> Here 'for', "in" and "else' are keywords.

=> Here iterable_object can be sequence (bytes, bytearray, range, str), list (list, tuple), set (set, frozenset) and dict.

=> The Execution process of for loop is that "Each of Element of iterable_object selected, placed in varname and executes indentation block of statements". This process will be repeated untill all elements of iterable_object completed.

=> When all the elements of iterable object completed then pvm comes out of for loop and executes else block of statements which are written under else block.

=> writing else block is optional.

```
#Char Display while loop
s="python"
i=0
while(i<len(s))
    print("\t{}".format(s[i]))
    i=i+1
```

Chardisplay forloop:-

s = "PYTHON" → s=input("Enter a word/line":")

s is called Iterable object

```
for k in s:
    print("\t{}".format(k))
```

mul table forloop:-

```
n=int(input("Enter a number of generating mul table:"))
if(n<=0):
    print("{} is invalid input".format(n))
else:
    print("_"*50)
    print("mul table for {}".format(n))
    print("-"*50)
    for i in range(1,11):
        print("\t {} x {} = {}".format(n,i,n*i))
    else:
        print("*"*50)
```

Factorial while

n! ----> if n is +ne — logic1 --> 1x2x3 x..n — logic2 --> n x (n-1) x (n-2)... 1

logic

if n is 0 = fact 1

if n is -ve = invalid input

3! ----> 1 x 2 x 3

```
f=1
i=1
while(i<=n):
    f=f*i
    i=i+1
else:
    print("fact=",f)
```

OR

```
f=1
for i in range(1,n+1):
    f=f*i
else:
    print("fact=",f)
```

3! => 3x2x1

```
f=1
i=n
while(i>0)
    f=f*i
    i=i-1
else:
    print("Fact=",f)
```

OR

```
f=1
for i in range(n,0-1):
    f=f*i
else:
    print("fact=",f)
```

fact while loop

```
n= int(input("Entr a numbe for cal factorial:"))
if(n<0):
    print("{} is invalid input". format(n))
else:
    f=1
    i=1
    while (i<=n):
        f = f*i
        i = i+1
    else:
        print("factorial({}) = {}".format(n,f))
```

Ex②.

```
n=int(input("Entr a number for cal factorial:"))
if (n < 0):
    print("{} is invalid input". format(n))
else:
    f=1
    i= n
    while (i>0):
        f = f* i
        i =i-1
    else:
        print ("factorial ({} same upr.
```

Fact for loop:-

```
else:
    f=1
    for i in range(1,n+1):
        f = f*i
    else:
        print("\t {}! = {}". format (n,f))
```

Ex③.

```
else:
    f=1
    for i in range (n, 0, -1):
        f=f*i
    else:
        print
```

#Factor

n=12

1 2 3 4 5 6 7 8 9 10 11 12

```
for i in range (1,n+1):
    if (n%i==0)
        print(i)
```

12/12 .

(OR)

```
for i in range (1,(n//2)+1):
    if(n%i==0)
        print (i)
```

Ex①

```
n=int(input("enter a number for getting its Factor:"))
if (n<=0):
    print ("{} is invalid input". format(n))

else:
    print("-"*50)
    print("factor of {}".format(n))
    print("-"*50)
    for i in range (1,n+1):
        if (n%i==0):
            print ("\t{}".format (i))
    else:
        print ("-"*50)
```

Ex②

```
    for i in range (1,(n//2)+1):
        if (n%i==0):
            print ("\t{}".format(i))

    else:
        print("\t{}".format (n))
        print ("-"*50)
```

program for finding sum of digits of a given number

Hint 3452
3+4+5+2
sum=14

```
n=3452
s=0
 if (n>0):
     r=n%10
     s = s+r
     n = n//10

n=345
if (n>0):
    r=n%10
    s=s+r
    n=n//10

n = 34
if (n>0):
    r=n%10
    s=s+r
    n = n/10

n=3
if (n>0):
    r=n%10
    s=s+r
    n =n//10
```

ex①

```
n=int(input("Enter a number:"))
if(n<=0):
     print("{} is invalid input".format(n))

else:
   s=0
   while (n>0):
        r=n%10
        s=s+r
        n = n//10
   else:
        print("Digits sum={}".format(s))
```

Ex②
```
if(int(n)<=0):
        print("{} is invalid input".format(n))
else:
    s=0
    print("Given number:{}".format(n))
    digits=n.split()
    for digit in digits[0]:
            s=s+int(digit)
    else:
        print("sum of digits:{}".format(s))
```

Ex③
```
digits=list(n)    # typecasting str into Digits.
for digit in digits:
    s=s+float(digit)
else:
    print("sum of digits={}".format(s))
```

for loop or for...else loop:-

- syntax 1:-
- for varname in iterable object:

  - indentation block of statements
- other statements in program
- syntax 2:-
- for varname in iterable object:

- - indentation of block of statement
- else:
- - else block of statements

- other statements in program

In [1]:
```
#char display while Loop
s="python"
i=0
while(i<len(s)):
    print("\t{}".format(s[i]))
    i=i+1
```

```
        p
        y
        t
        h
        o
        n
```

In [3]:
```
#char display for loop
s=str(input("enter word or line"))
for k in s:
    print("\t{}".format(k))
```

```
        m
        d

        a
        r
        i
        f

        r
        a
        z
        a
```

In [4]:
```
# multable for loop:
n=int(input("enter a number of generating mul table:"))
if(n<=0):
    print("{} is invalid input".format(n))
else:
    print("="*50)
    print("multable for {}".format(n))
    print("="*50)
    for i in range(1,11):
        print("\t{}*{}={}".format(n,i,n*i))
    else:
        print("="*50)
```

```
==================================================
multable for 5
==================================================
        5*1=5
        5*2=10
        5*3=15
        5*4=20
        5*5=25
        5*6=30
        5*7=35
        5*8=40
        5*9=45
        5*10=50
==================================================
```

In [13]:
```
#factorial while loop
#logic1
n=int(input("enter  number for calculating factorial:"))
if(n<=0):
    print("\t{} is invalid input".format(n))
else:
    f=1
    i=1
```

```
    while(i<=n):
        f=f*i
        i=i+1
    else:
        print("factorial({}!)={}".format(n,f))
```

factorial(4!)=24

In [12]:
```
#factorial while Loop
#logic2
n=int(input("enter a number for calculating factorial:"))
if(n<=0):
    print("\t{} is invalid input".format(n))
else:
    f=1
    i=n
    while(i>0):
        f=f*i
        i=i-1
    else:
        print("factorial ({}!)={}".format(n,f))
```

factorial (5!)=120

In [21]:
```
#factorial for  for Loop:
#logic1
n=int(input("enter a number for calculating factorial:"))
if(n<=0):
    print("\t{} is invalid input".format(n))
else:
    f=1
    for i in range(1,n+1):
        f=f*i
    else:
        print("\t{}!={}".format(n,f))
```

	8!=40320

In [19]:
```
#factorial for  for Loop:
#logic2
n=int(input("enter a number for calculating factorial:"))
if(n<=0):
    print("\t{} is invalid input".format(n))
else:
    f=1
    for i in range(n,0,-1):
        f=f*i
    else:
        print("\t{}!={}".format(n,f))
```

	4!=24

In [29]:
```
#factors
#logic1
n=int(input("enter a number of getting its factors:"))
if(n<=0):
    print("{} is inavalid input:".format(n))
else:
    print("="*50)
    print("factor of {}".format(n))
    print("="*50)
    for i in range(1,n+1):
```

```
        if(n%i==0):
            print("\t{}".format(i))
        else:
            print("="*50)
```

```
==================================================
factor of 4
==================================================
	1
	2
==================================================
	4
```

In [28]:

```
#factors
#logic2
n=int(input("enter a number of getting its factors:"))
if(n<=0):
    print("{} is inavalid input:".format(n))
else:
    print("="*50)
    print("factor of {}".format(n))
    print("="*50)
    for i in range(1,(n//2)+1):
        if(n%i==0):
            print("\t{}".format(i))
        else:
            print("="*50)
```

```
==================================================
factor of 4
==================================================
	1
	2
```

In [35]:

```
#program for finding sum of digits of a given numbers:
#logic1
n=int(input("enter a number:"))
if (n<=0):
    print("{} is invalid input".format(n))
else:
    s=0
    while(n>0):
        r=n%10
        s=s+r
        n=n//10
    else:
        print("digits sum={}".format(s))
```

```
digits sum=39
```

In [42]:

```
#program for finding sum of digits of a given numbers:
#logic2
n=input("enter a number:")
if (int(n)<=0):
    print("{} is invalid input".format(n))
else:
    s=0
    print("given number :{}".format(n))
    digits=n.split()
    for digit in digits[0]:
        s=s+int(digit)
```

```
    else:
        print("sum of digits={}".format(s))
```

```
given number :547
sum of digits=16
```

In [43]:

```
#program for finding sum of digits of a given numbers:
#logic3
n=input("enter a number:")
if (int(n)<=0):
    print("{} is invalid input".format(n))
else:
    s=0
    print("given number :{}".format(n))
    digits=list(n)       #typecasting str into digits
    for digit in digits:
        s=s+float(digit)
    else:
        print("sum of digit ={}".format(s))
```

```
given number :5478
sum of digit =24.0
```

In [ ]: